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Superimposed Recent, Permo-Carboniferous and Ordovician palaeomagnetic remanence in the Builth Volcanic Series, Wales

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The Builth Volcanic Series of Llanvirnian age in Llanelwedd Quarries, mid-Wales, carries three components of natural remanent magnetisation. Component P, regarded as primary, is a thermochemical remanence directed at $D = 181.7°$, $I = +54.5°$, $\alpha_{95} = 4.4°$ relative to bedding. Component S is a secondary component with in situ $D = 178.7°$, $I = -6.7°$, $\alpha_{95} = 5.4°$ and is believed to be a low-temperature chemical remanence (CRM) of Permo-Carboniferous age. Component R is directed close to the present geomagnetic field and is believed to be a recent viscous remanence (VRM).

The results are of interest for three reasons. First, they are an unusually good example of multi-component NRM analysis, the three components being so clearly discriminated by thermal demagnetisation because they have almost completely separate blocking temperature ranges. Second, they provide evidence of a Permo-Carboniferous event (possibly a mild thermal or hydrothermal pulse promoting CRM acquisition) some 40 km north of the Hercynian orogenic front. Third, they illustrate very clearly the importance of detailed demagnetisation: this work revises the pole position for these rocks by ~ 10° and removes an obstacle to the palaeomagnetic recognition of the ~ 1000 km wide Iapetus Ocean cutting Britain in Ordovician time.

1. Introduction

Palaeomagnetic directions and pole positions from Ordovician rocks from the British Isles show differences which have variously been attributed to local tectonic complexity, erratic apparent polar wandering and failure to average out palaeosecular variation. No clear pattern is discernible among these data in terms of apparent polar wander or regional difference which would imply relative (micro)plate motion except that inclinations of stable NRM, and hence palaeolatitudes, seem generally to be lower for Britain north of the Iapetus suture than for southern Britain [1,2]. Some individual datasets, however, conflict with this generalisation and therefore inhibit one from confidently equating the difference in palaeolatitude with the minimum width of the Iapetus Ocean in Ordovician time.

One such conflicting piece of data arises from the Builth Volcanic Series [3,4]. Those data were derived almost entirely from alternating field demagnetisation with only very limited thermal demagnetisation [4]. The quarry section studied by previous workers [3,4] has now been resampled and the new collection has been subjected to detailed thermal demagnetisation, together with a.f. treatment of a single specimen from each stratigraphic level for comparative purposes, and the results are reported here.

The Builth Volcanic Series is best exposed in the Llanelwedd Quarries (NGR SO 04805222-05155213) at the southern end of an inlier of some 15 km × 18 km of Ordovician rocks between Builth Wells and Llandridnod Wells, mid-Wales (~ 52.2° N, ~ 3.3° W). It was mapped by Jones and Pugh [5] upon whose work the recent IGS 1 : 25,000

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map is based [6]. The Wenlockian sediments which surround the inlier are only gently dipping; hence the Ordovician rocks that are the subject of the present study had essentially acquired their present attitude before mid-Silurian time. The Builth Volcanic Series is a submarine spilite-keratophyre assemblage, subjected to soda metasomatism during its accumulation. Its setting was near the southeast margin of the Ordovician Iapetus Ocean. The sequence studied here strikes ~ 160° and dips ~ 35° W, the deviation from the usual southwest Caledonian strike being presumably due to its position on a large-scale synclinal structure of which several occur within the inlier.

The same 400 m sequence studied by Piper and Briden [4] has been resampled in this new work, with the addition of clasts in an agglomerate (stratigraphically between their sites 6 and 7) which give valuable control on the age of the various components of remanence. The stratigraphic relation between the individual sites in the old and new studies is only approximately known. Five horizons are sampled in the spilite and keratophyre lavas of the upper part of the section covered by Piper and Briden's [4] sites 7–15. We sampled at least six levels in the lower part of the section covered by their sites 1–6, but it is not certain that our sampling extended as far down the sequence as the earlier work. Also, it is possible that more of this lower part of the section consists of lavas rather than tuffs as previously inferred [5,4].

2. Thermal demagnetisation analysis

All the thermal demagnetisation behaviour can be described in terms of three components of NRM which have the following blocking temperature (T_B) and in situ directional characteristics:

Component R: $T_B \lesssim 300$ °C and mostly $\lesssim$ 200 °C, directed north and downward, although the direction is not well determined and may vary substantially from specimen to specimen.

Component S: T_B typically 200–500 °C with little if any blocking temperatures outside this range, directed southward and with very shallow, usually slightly upward, inclination.

Component P: T_B almost entirely in the range 500–590 °C with occasional evidence of both lower and higher T_B; it is directed SSE and downward.

The whole range of behaviour is illustrated by the orthogonal projections in Fig. 1; all three components are present in almost all specimens, and plot as linear segments in the projections. In Fig. 1a and b they are present in about equal abundance; in Fig. 1c P predominates with a little R and no S; Fig. 1d is similar, but with a small amount of S; in Fig. 1e S predominates, but the other two components are clearly present; in Fig. 1f and g R and S predominate, but P is still well defined. Overlap of blocking temperatures is rare; indeed the only clear example is in Fig. 3c in which curvature of the trajectory from 475 to 575 °C indicates that a small fraction of S has blocking temperatures in the range dominated by P (but this sample comes from an agglomerate clast and is discussed below).

Best estimates of the directions of the various components are calculated by the weighted principal component analysis routine of Kent et al. [7] which also produces the blocking temperature spectra that are illustrated as insets in Fig. 1.

The directional distribution of the components P, S and R is illustrated in Fig. 2, both in situ and with the bedding corrected by simple rotation about the present strike as an estimate of Ordovician coordinates, and the statistics of their distributions are given in Table 1. Because of the limited range of variation of dip across the section, the differences in dispersion in Present and Ordovician reference frames are insufficient to indicate the ages of each component relative to the tilting. However, this information can be deduced from the thermal demagnetisation data from the clasts in the agglomerate, and by reference to the known apparent polar wander path for England and Wales.

The andesitic clasts in the agglomerate have three components of NRM, designated P_{ag}, S_{ag} and R_{ag}, with T_B ranges corresponding to P, S and R in the main sequence (Fig. 3, cf. Fig. 1). Both R_{ag} and S_{ag} are similar in direction and precision (k) to R and S in the rest of the sequence (Fig. 4 and Table 1). Component P_{ag} by contrast is not significantly grouped ($p = 0.05$) (Table 1) in the clasts. Hence at least some of component P_{ag} in the clasts

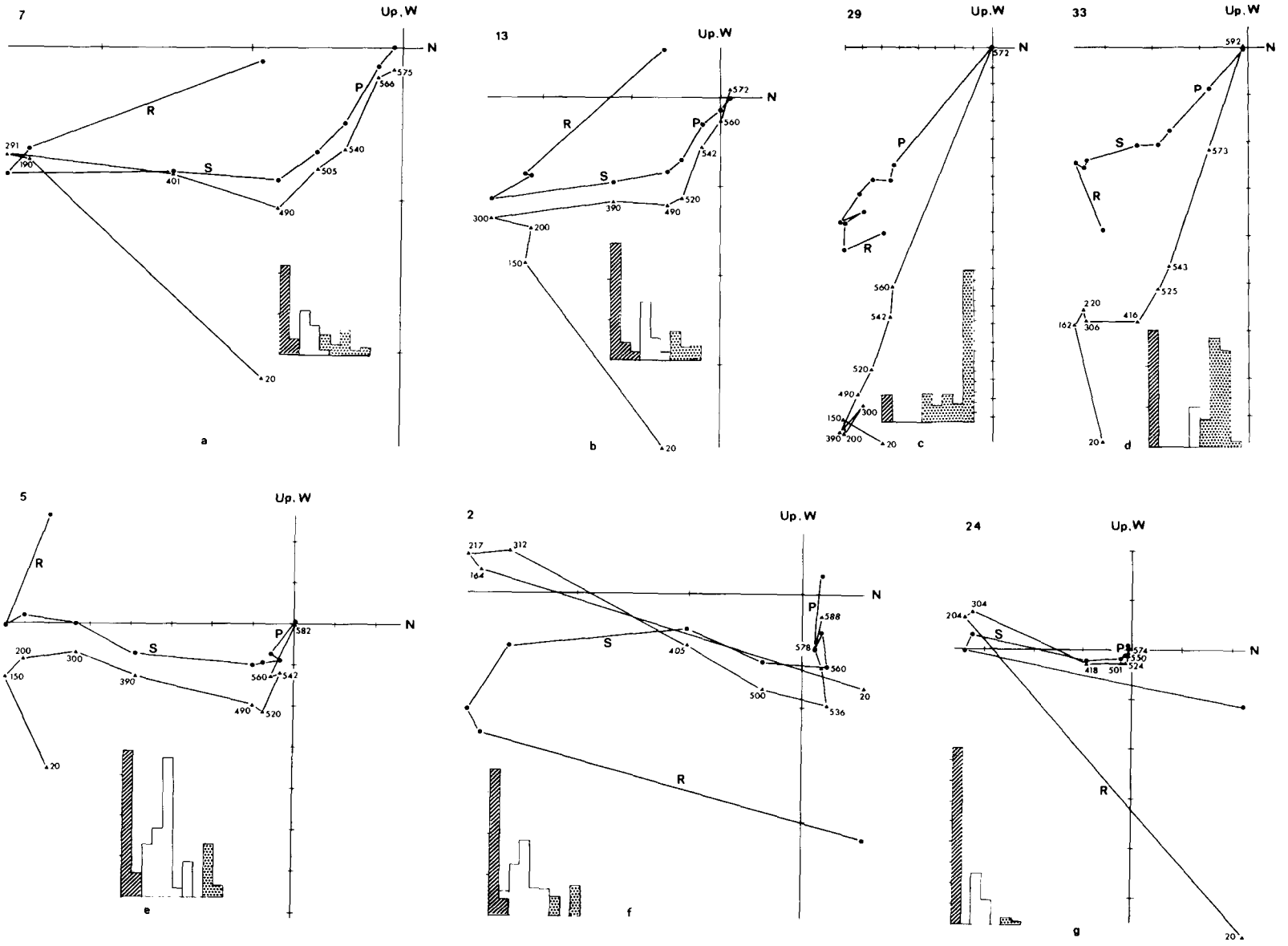


Fig. 1. Thermal demagnetisation of typical samples: orthogonal projections of in situ NRM directions on to the horizontal (●) and N-S vertical (▲) planes. The temperatures nominally reached are indicated alongside the vertical projection, and the interpretation of the components being removed at different stages is indicated alongside the horizontal projection. Sample number is given in the top left-hand corner. The blocking temperature spectrum (inset) is estimated by modelling the remanence in terms of the components R (shaded), S (blank) and P (stippled)—see text—and determining the amount of each that is removed in each demagnetisation step (hence the abscissa of these insets is not linear in temperature). The pecked marks on the axes are at intervals of 1 mA m^{-1}.

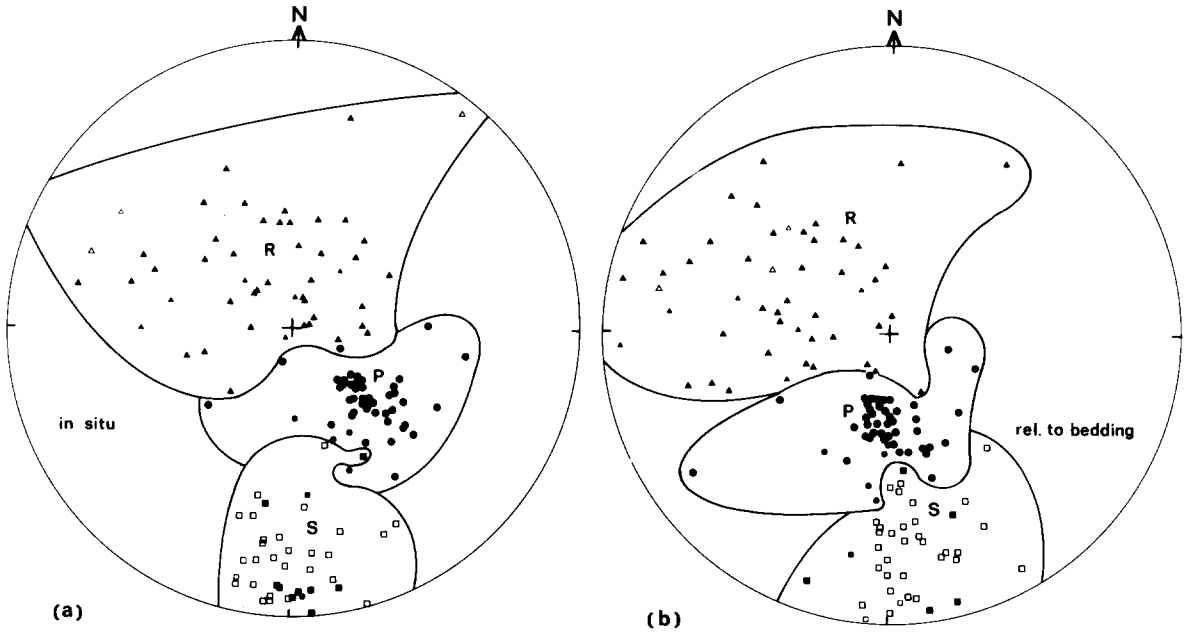


Fig. 2. NRM components, identified from analysis of linear structure in demagnetisation trajectories. Low blocking temperature T_B/low coercivity H_{CR} denoted by triangles labelled "R"; intermediate T_B or H_{CR} denoted by squares and labelled "S"; high T_B or H_{CR} denoted by circles and labelled "P". Open symbols denote upward (negative inclination) and solid symbols downward (positive inclination). The a.f. demagnetisation data are depicted with smaller symbols. Stereographic projection. (a) In situ, (b) dip corrected by simple rotation about the present strike.

TABLE 1

Fisher statistics of the NRM components in the Builth Volcanic Series

Main Bedded Sequence

Component	N	In situ					Dip corrected				
		R	k	$\alpha_{95}(°)$	$D(°)$	$I(°)$	R	k	$\alpha_{95}(°)$	$D(°)$	$I(°)$
R	49	38.8048	4.7	10.5	336.7	+64.6	39.1322	4.8	10.3	291.9	+44.5
S	39	37.0067	19.1	5.4	178.7	−06.7	36.9664	18.6	5.4	173.3	−12.3
P	49	46.9736	23.6	4.2	134.3	+50.0	46.7787	21.6	4.4	181.7	+54.5

Agglomerate

Component	N	R	k	$\alpha_{95}(°)$	In situ		Dip corrected	
					$D(°)$	$I(°)$	$D(°)$	$I(°)$
R	6	5.4030	8.4	24.6	310.3	+46.1	295.2	+26.3
S	4	3.8293	17.5	22.5	165.1	+03.0	167.3	+04.2
P	6	3.6706	–	–	–	–	–	–

The statistical notation in the headings of this table are conventional [8]; N denotes the number of samples in which each component is identified. When this analysis is done on the basis of unit weight for each stratigraphic level the mean direction estimates are identical to better than 1°, but the precision estimates are higher, being about 8, 76 and 47 for R, S and P respectively, observed at 11, 9 and 11 horizons respectively.

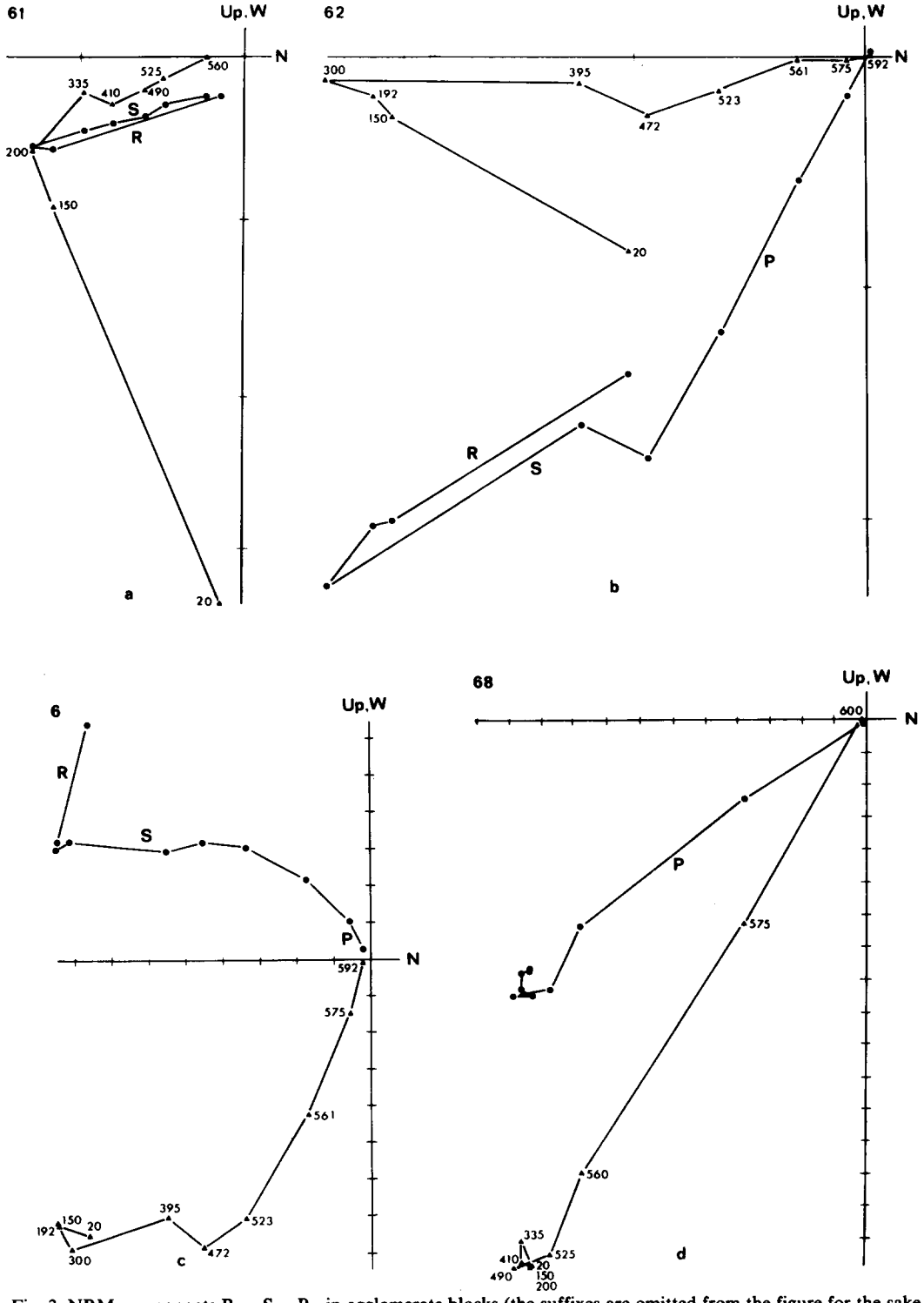


Fig. 3. NRM components R_{ag}, S_{ag}, P_{ag} in agglomerate blocks (the suffixes are omitted from the figure for the sake of design): in situ orthogonal projects; for notation see Fig. 1.

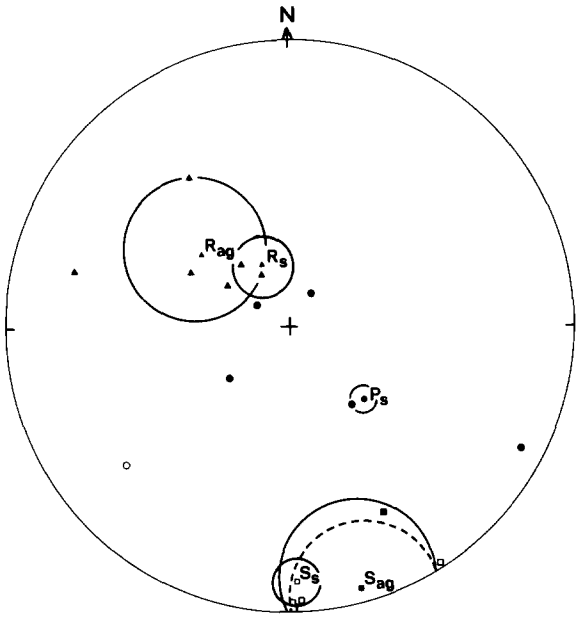


Fig. 4. Stereographic projection showing the in situ NRM components in the agglomerate blocks (symbols as in Fig. 2), together with the mean (smaller symbols) and circle of 95% confidence of R_{ag} and S_{ag}, and of R_s, S_s and P_s in the main volcanic sequence.

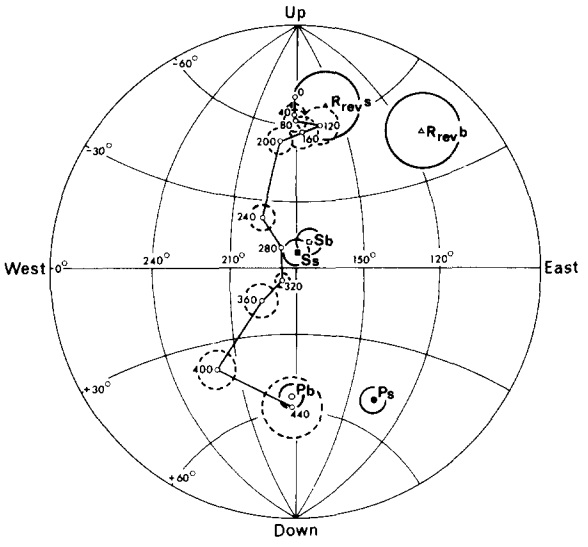


Fig. 5. Remanence components in the Builth Volcanic Series compared with the expected directions of the reversed geomagnetic field at Builth Wells from Ordovician to the present time. The reverse of component R is plotted (triangles), S is denoted by squares and P by circles; subscripts "s" and solid symbols distinguish in situ directions, and open symbols and subscripts "b" denote directions relative to bedding. Circles of 95% confidence are also shown. The expected directions are calculated from the global palaeomagnetic data and continental reassemblies of Smith et al. [9] at 40 m.y. intervals back to 360 m.y.; the 400 and 440 m.y. directions were inferred from other data from the British Isles only [1]. Stereographic projection centred on $D = 180°$, $I = 0°$.

originated prior to agglomerate formation and is therefore syngenetic with the accumulation of the volcanic pile; the component designated P in the bedded sequence has the same T_B characteristics and is therefore regarded as originating in the same way at the same time.

Since P therefore predates the tectonic tilting of these rocks it is not surprising that its in situ direction does not correspond to any known Ordovician or younger palaeomagnetic field for England and Wales (Fig. 5). The "dip corrected" direction of P, however, agrees very closely with previous estimates of the Ordovician palaeomagnetic field in this region, and is a valid estimate of the local field in Llanvirn time (Fig. 5, Table 1). We chose the notation P to denote "primary".

R and S, by contrast, postdate agglomerate formation and, in "dip corrected" coordinates, correspond to no known Ordovician or younger palaeomagnetic field for England and Wales (Fig. 5). They are likely to post-date the tilting. In situ, S corresponds to the local Permo-Carboniferous field: it dates from the end of the Hercynian orogeny. We chose the notation S to denote "secondary".

The lowest blocking temperature component, is statistically indistinguishable ($p = 0.05$) from the present local geomagnetic field (Fig. 4). It is clearly a Recent VRM hence our choice of designation R.

The origin of S is less clear. The Builth Inlier is 40 km north of the Hercynian fold front, and is surrounded by Silurian sediments which are not recognisably metamorphosed and are unlikely to have been heated above 200 °C by any post-emplacement metamorphism or burial. The maximum blocking temperature of S is about 500 °C; even allowing for effects of being held for long periods of time at low metamorphic temperatures, there is no way that such blocking temperatures can be

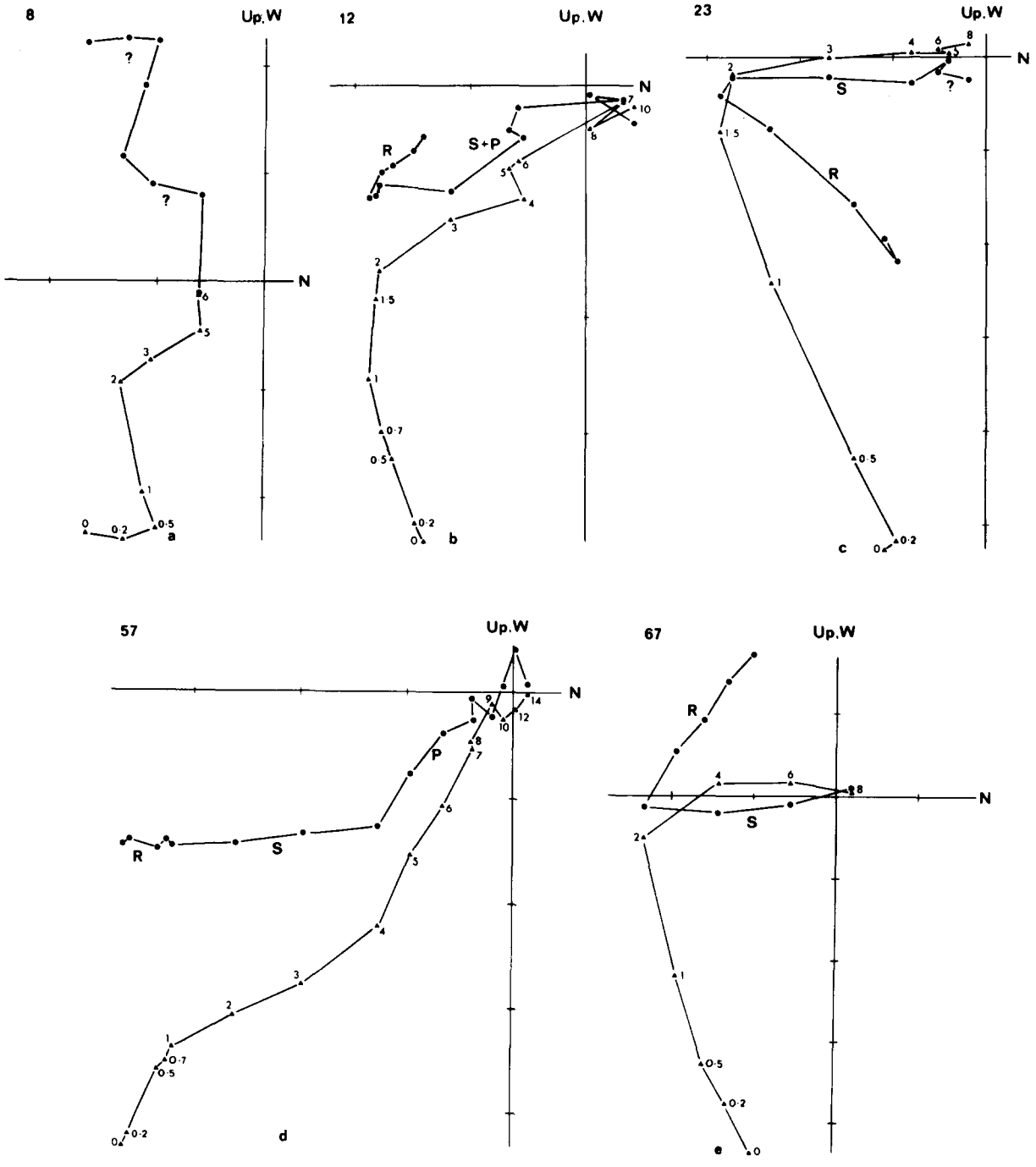


Fig. 6. Typical alternating field demagnetisation behaviour. Notation as Fig. 1 except that demagnetisation treatment is in units of 10 mT and (for sample 57, in (d), only) the scale is in units of 10 mA m^{-1}.

explained as a PTRM. S must be at least mostly a CRM, but because it is so well defined (Fig. 3) it must have been acquired in a single event, and it seems likely that the CRM process was accelerated by a mild thermal pulse.

At first thought, the foregoing argument might be taken as evidence that component P is a pure TRM. However, the resultant of P in the conglomerate is directed close to P in the bedded sequence, and although it fails a significance test at $p = 0.05$, it passes at $p = 0.10$: hence it seems that the magnetisation process within the clasts might still have been proceeding during or after the agglomerate formation and diagenetic CRM may be at least a contributor to the origin of P. It seems likely to be due to the metasomatism of the volcanic pile during its accumulation.

3. A.f. demagnetisation analysis

A single sample from eight of the stratigraphic horizons was subjected to a.f. demagnetisation, using the same equipment as Piper and Briden [4]. Only one of these (Fig. 6d) showed three distinct components P, S, R. In most other cases R was identified as a low-coercivity component, but at higher a.f. steps the vector removed was either S (Fig. 6c) or intermediate between P and S (compare Fig. 6b with the distinct P and S in Fig. 1b from the same site). Sample 08 demagnetised very erratically in alternating fields (Fig. 6a) whereas its sister sample 07 (Fig. 1a) showed the three components P, S, R very clearly on thermal demagnetisation. A.F. demagnetisation of an agglomerate sample 67 (Fig. 6e) showed R and S only, rather like 23 in the bedded sequence (Fig. 6c) and also like thermal demagnetisation of sample 61 from the agglomerate (Fig. 3a).

With the benefit of hindsight, the results described in the preceding sections could perhaps have been anticipated from Piper and Briden's [4] results. The presence of the soft steep northerly component in their data is indicated by the southerly movement of site mean directions after a.f. cleaning [4, fig. 2] and the existence of a shallow southerly component is hinted at by the planar distribution of "cleaned" site mean directions due to the presence of a few shallow southerly directions. Their progressive demagnetisation data hint that there are three NRM components in some samples insofar as both a.f. and thermal demagnetisation at sites 6 and 7 [4, fig. 4] shows V-shaped direction change, first moving southward toward a shallow direction, then steepening. The absence of any directional change in some samples with moderately steep southerly NRM indicates that this is the direction of the most stable component, and that the other components are not ubiquitous throughout the collection.

4. Conclusions

The hardest NRM component in the Builth Volcanic Series is syngenetic with the formation of the volcanic pile in Llanvirnian times and could have been formed by a combination of TRM and CRM mechanisms. Its direction is $D = 181.7$, $I = +54.5$, $\alpha_{95} = 4.4$. The inclination is steeper and hence the inferred palaeolatitude is more southerly than had been inferred by previous workers. The palaeomagnetic pole is $\sim 10°$ different from that in earlier studies. It is now located near the equator (2.8°S, 4.8°W; $d\psi = 4.4°$, $d\psi = 6.3°$), together with other palaeomagnetic poles from Ordovician rocks in England and Wales, instead of being located around 12°S among Ordovician poles from northern Britain [10]. The discrepancy between those two groups can be interpreted as indicating the existence of an ocean about 1000 km wide [10] between northern and southern Britain in Ordovician times: the Iapetus Ocean.

The near-horizontal southerly component of NRM is of Permo-Carboniferous age, and is a low-temperature CRM, possibly activated by a mild thermal pulse associated with the Hercynian orogeny. The identification of such a distinct event 40 km north of the Hercynian orogenic front is noteworthy. The declination of this component is slightly less than that expected by reference to other studies in Europe and could possibly indicate a counter-clockwise rotation of the study area by 10° or so since Permo-Carboniferous time—an adjustment which would also apply to the Ordovician palaeomagnetic directions. However, there is

no corroborative evidence of such a rotation and it is perhaps more likely that this small discrepancy is due to the inadequacy of the reference curve in Fig. 5.

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